

Time Emergence: FFGFT and GALG Compared

Decompactification vs. Co-exclusion —
Two Algebraic Paths to Observable Time

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Time Emergence and Gravitational Constant: Two Suggestions for Matzke's GALG Framework

Abstract

This document offers two suggestions for Doug Matzke's GALG framework [1, 7], supported by direct quotations and algebraic verification.

Proposal 1 (time): Matzke's co-exclusion primitive generates proto-time through operator change [1] (p. 37), in direct parallel to FFGFT decompactification (unrolling, Doc. 333). Both frameworks share the same three-layer emergence structure. This is not a weakness of GALG but an unspoken parallel that strengthens it.

Proposal 2 (G): The Schwarzschild derivation in [7] starts from the hyperbit $e_i^2 = +1$ — a purely algebraic, G-free initial condition. The SI interface is introduced only through T_{Planck} , from which, with $T = T_{\text{Planck}}/(2\pi)$, the value $M_{\text{bit}} = m_P \cdot \ln 2/(2\pi)$ follows. Since $m_P = \sqrt{\hbar c/G}$, this interface contains G — but this is structurally unavoidable: any theory that produces SI units must build a bridge to measured constants somewhere. T_{Planck} is that bridge, and it is legitimate. The suggestion: restrict the claim "*without free parameters*" to the algebraic layer $e_i^2 = +1 \rightarrow \text{Cl}(6)$, and name T_{Planck} explicitly as the SI interface. In FFGFT, ξ and m_e serve as the SI interface (Doc. 012) — a different route, but the same structural necessity.

Both proposals are documented through direct quotations from Matzke's publications and verified by Python scripts.

Guide

The basic problem. Observable time is a macroscopic phenomenon. Quantum mechanics and algebraic field theories operate at a level where time is either absent or appears only as a discrete primitive. How does continuous, observable time arise from a timeless algebraic base structure?

What this document shows. Two suggestions, supported by direct quotations from his own publications: (1) Time emergence in GALG has the same structure as in FFGFT — this strengthens Doug's framework but has not yet been made explicit. (2) The G derivation in [7] is algebraically necessary from the SI interface T_{Planck} onward — the claim “*without free parameters*” holds precisely for the algebraic layer, not for the SI embedding.

Notation

Symbol	Meaning	Source
<i>Status markers</i>		
[B]	algebraically proved, script available	Doc. 190
[K]	derivable from ξ	Doc. 190
[S]	open bridge, sketch	Doc. 190
<i>FFGFT</i>		
$\tilde{T} = 1/m$	intrinsic time scale of a mode	Doc. 321 [4]
$T^4/\mathbb{Z}_3$	compact 4-torus geometry	Doc. 257 [3]
Decompactification	unrolling of the compact dimension	Doc. 333 [5]
<i>Matzke GALG</i>		
Co-occurrence	addition $a + b$: spatial primitive	[1] p. 37
Co-exclusion	multiplication ab : temporal primitive	[1] p. 37
Proto-time	emergent time order from co-exclusion	[1] p. 14

.1 Time in FFGFT: compactified and emergent

The compactified time scale $\tilde{T} = 1/m$

In FFGFT, each particle of mass m carries an intrinsic time scale (Doc. 321 [4]):

$$\tilde{T} = \frac{1}{m} \quad (\text{natural units, } \hbar = c = 1) \quad (1)$$

This time scale lives on the compact geometry $T^4/\mathbb{Z}_3$. It is mode-specific — each mode has its own time scale, and cross-mode superpositions generate incompatible time scales (Doc. 334 [6]) **[B]**.

Decompactification as the emergence mechanism

The compact dimension is not directly observable. Observable time arises only through unrolling: the transition from compact geometry to macroscopic spacetime (Doc. 333 [5]).

Doc. 333 distinguishes two unrolling types:

- **Type I** (measurement act): lossy, irreversible — wavefunction collapse as decompactification.
- **Type III** (pullback): lossless, reversible — geometric retraction to the compact manifold.

Observable time is a Type-I phenomenon: it arises through the irreversible unrolling act, not from the compact geometry itself.

Core statement FFGFT [B]: Before decompactification, only $\tilde{T} = 1/m$ exists as a mode-bound geometric quantity. Classical, measurable time t is the result of unrolling.

.2 Time in GALG: proto-temporal and emergent

Co-occurrence and co-exclusion

Matzke [1] defines two fundamental primitives for geometric algebra (Ch. 4.2, pp. 36–39):

Co-occurrence (spatial primitive): two states exist simultaneously, expressed as GA addition $a + b$. Direct quote [1], p. 37:

“Co-occurrence is defined as two or more states ‘occurring’ simultaneously [...] commutative properties of addition reflects the lack of temporal ordering characteristic of simultaneity.”

Co-occurrence is static — no temporal ordering.

Co-exclusion (temporal primitive): an operator changes the state through GA multiplication ab . Direct quote [1], p. 37:

“Since co-exclusion infers an operator for change and implies temporal sequence, it is considered to be the temporal primitive. Conversely, co-occurrence is labeled the spatial primitive since it is static.”

Proto-time emerges from operators

Classical time is not encoded in the GALG framework, but emerges from the interaction of disjoint operators. Direct quote [1], p. 14:

“Interaction of disjoint finite sets of dimensions (operators and resulting co-exclusions) creates a synchronization-based proto-time from which classical time and energy metrics ultimately emerge.”

On the need for a different temporal concept in the quantum domain (p. 12):

“The quantum domain must have a different kind of primitive temporal framework, if for no other reason than it exists in a completely different spatial environment. Conceptually, this proto-temporal framework is identical to the synchronization tokens and mechanisms used by asynchronous logic.”

Core statement GALG: Before emergence, only the algebraic primitive of co-exclusion (multiplication) exists. Classical time and energy arise from the interaction of disjoint operator sets.

.3 The structural parallel

Layer	FFGFT	GALG
Timeless base layer	$\tilde{T} = 1/m$ compactified	co-occurrence (addition)
Temporal primitive	torsion / winding on T^4	co-exclusion (multiplication ab)
Emergence mechanism	decompactification (unrolling)	interaction of disjoint operators
Result	observable time t , mode-bound	classical time + energy

Both frameworks share the same three-layer structure: (1) a timeless algebraic base layer, (2) a temporal primitive that implies sequence without being continuous itself, (3) an emergence mechanism that produces classical, measurable time.

.4 Open question

Is Matzke's co-exclusion the algebraic expression of the same structure that FFGFT describes as decompactification?

Concretely: in FFGFT, unrolling Type I (measurement act, collapse) is an irreversible transition — just as co-exclusion describes a state change through an operator without possibility of reversal (Matzke [1], p. 145: *"The process is irreversible because a multiplicative inverse does not exist for $(1 \pm a_0)$ and $(1 \pm a_1)$ "*).

This parallel — irreversibility as a shared feature — could be the algebraic core of the connection [S].

Verification would require:

1. Explicit construction of the unrolling operator in GALG notation.
2. Proof that co-exclusion + irreducibility reproduces the FFGFT mass formula $\tilde{T} = 1/m$.

.5 Mass and gravitational constant: derived or implicit?

Matzke's Schwarzschild derivation and the implicit G

In his IPI Letters paper 2026 [7], Matzke presents the derivation as follows:

"This paper derives the complete black hole picture — Schwarzschild radius, Bekenstein entropy, Hawking radiation, collapse threshold, and information conservation — from a single algebraic axiom with no general relativity and no free parameters."

The derivation uses $M_{\text{bit}} = m_p \cdot \ln 2 / (2\pi)$ as starting point (Landauer at $T_{\text{Planck}} / (2\pi)$, then $E = mc^2$). From the collapse condition Compton = Schwarzschild:

$$\frac{\hbar}{M_{\text{coll}} c} = \frac{2GM_{\text{coll}}}{c^2} \Rightarrow G = \frac{\hbar c}{2 M_{\text{coll}}^2} \quad (2)$$

with $M_{\text{coll}} = n_{\text{thresh}} \cdot M_{\text{bit}} = m_p / \sqrt{2}$ (numerically verified: 0.00% deviation).

The implicit G from this construction:

$$G_{\text{doug}} = \frac{\hbar c}{2 M_{\text{coll}}^2} = \frac{\hbar c}{m_p^2} = 6.6743 \times 10^{-11} \text{ m}^3 / (\text{kg} \cdot \text{s}^2) \quad (3)$$

Deviation from experimental value: 0.00 %.

The SI interface and what it achieves

Matzke's initial condition is purely algebraic and G-free:

$$e_i^2 = +1 \quad (\text{hyperbit, } +1 \text{ signature}) \quad (4)$$

From this follows the Witt decomposition into nilpotents $\{ap, an\}$ with $ap^2 = 0$, then Cl(6), then the Planck voxel — all without G.

The SI interface enters only through T_{Planck} :

$$M_{\text{bit}} = \frac{k_B (T_P/2\pi) \ln 2}{c^2} = \frac{m_P \ln 2}{2\pi} \quad (5)$$

Only with this value does $n_{\text{thresh}} \cdot M_{\text{bit}} = m_P/\sqrt{2}$ hold, with $n_{\text{thresh}} = 2\pi/(\sqrt{2} \ln 2) \approx 6.41$ (not an integer); with $k_B T_P \ln 2/c^2 = m_P \ln 2$ one would obtain $2\pi/\sqrt{2} \cdot m_P = 4.44 m_P$. The starting value thus requires the Landauer temperature $T = T_P/(2\pi)$. Since $m_P = \sqrt{\hbar c/G}$, this interface contains G — but this is structurally **unavoidable**: any theory producing dimensionful SI quantities must insert measured constants somewhere. T_{Planck} is the minimal such bridge.

Matzke himself names this clearly in his IPI mail of 27 Aug 2026:

“the fact that M_{bit} is directly related to M_{Planck} was a happy coincident based on Planck units consolidation only.”

The suggestion: restrict the claim “*without free parameters*” explicitly to the algebraic layer $e_i^2 = +1 \rightarrow \text{Cl}(6)$. The SI embedding via T_{Planck} is legitimate and necessary — but should be named as such, not presented as part of the parameter-free derivation.

FFGFT: G as genuine derivation

In FFGFT, G is derived from ξ and m_e (Doc. 012 [8]), *without* m_P as input:

$$G = \frac{\xi^2}{4 m_e} \cdot C_{\text{conv}} \cdot K_{\text{frak}} \quad \text{with } \xi = \frac{4}{30\,000}, K_{\text{frak}} = \frac{74}{75} \quad (6)$$

with $C_{\text{conv}} = 7.783 \times 10^{-3} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2} \text{ MeV}$ and $m_e = 0.511 \text{ MeV}$; this gives $G = 6.679 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ (+0.07 %; with $K_{\text{frak}} = 0.986$: 6.6745×10^{-11} , +0.003 %).

Here ξ is fixed by the $T^4/\mathbb{Z}_3$ geometry and m_e by the mass formula, both independent of G . This is a genuine derivation: G is output, not input. This holds for the form of G and its order of magnitude, which follow from ξ and the single anchor; C_{conv} is the SI conversion of the single-anchor chain (Doc. 383), and the agreement with the CODATA value to the stated accuracy arises only with the small residual of this chain that is contained in C_{conv} . It is therefore not an independent proof of precision.

Comparison

Quantity	GALG (Matzke)	FFGFT
Input	m_P (contains G !)	ξ, m_e
Mass	$M_{\text{bit}} = m_P \cdot \ln 2 / (2\pi)$	$\tilde{T} = 1/m$, then ξ
Gravitation	$G = \hbar c / m_P^2$ (tautology)	$G = \xi^2 C_{\text{conv}} / (4m_e) \cdot K$
Status	algebraically consistent [S]	genuine derivation [K]

Both frameworks recover the measured value of G (FFGFT: +0.07 % with $K_{\text{frak}} = 74/75$, +0.003 % with $K_{\text{frak}} = 0.986$, the last digit via the residual of the single-anchor chain in C_{conv}), and both require an SI interface. GALG chooses T_{Planck} — minimal and physically well motivated. FFGFT chooses ξ and m_e — derived geometrically from the torus. Both routes are legitimate; the difference is the choice of bridge, not whether a bridge is needed.

Note on AI assistance

This document was produced with AI-assisted support (Claude, Anthropic). All direct quotations were taken from Matzke’s PhD [1] and verified with page references. The author bears sole responsibility for the content. Methodological principles: A-series, Doc. 190.

References

Bibliography

- [1] D. J. Matzke, *Quantum Computation Using Geometric Algebra*, Ph.D. Dissertation, University of Texas at Dallas, 2002. https://www.matzkefamily.net/doug/papers/PHD/phd_matzke_final.pdf
Direct quotes in this doc.: p. 12 (proto-temporal framework), p. 14 (proto-time from co-exclusion), p. 37 (temporal primitive), p. 145 (irreversibility).
- [2] D. J. Matzke, *GALG: Geometric Algebra over $\mathbb{Z}_3\mathbb{C}$* , IPI Talk, 26 July 2025. <https://informationphysicsinstitute.blogspot.com/2025/07/ipi-talk-doug-matzke-wwwquantumdougcom.html>
- [3] J. Pascher, *Doc. 257: $T^4/\mathbb{Z}_3$ Geometry and FFGFT*, FFGFT Corpus, ORCID 0009-0000-6518-4064. <https://github.com/jpascher/T0-Time-Mass-Duality>
- [4] J. Pascher, *Doc. 321: $SU(3)$ and $\mathbb{Z}_3$ Emergence; intrinsic time scale $\tilde{T} = 1/m$* , FFGFT Corpus, ORCID 0009-0000-6518-4064.
- [5] J. Pascher, *Doc. 333: K_{frak} , Unrolling and the $\tilde{T} \cdot m$ Duality*, FFGFT Corpus, 27 August 2026, ORCID 0009-0000-6518-4064.
- [6] J. Pascher, *Doc. 334: Superposition without Time*, FFGFT Corpus, 27 August 2026, ORCID 0009-0000-6518-4064.
- [7] D. J. Matzke, *Black Holes from Hyperbits: The Event Horizon as Nilpotent Algebra, Planck Voxels, and High-Dimensional Corner Geometry*, IPI Letters 2026, Vol. 4(3): N1-N22. <https://ipipublishing.org/index.php/ipil/article/download/378/201/1385>
- [8] J. Pascher, *Doc. 012: Derivation of the gravitational constant G from ξ* , FFGFT Corpus, ORCID 0009-0000-6518-4064. <https://github.com/jpascher/T0-Time-Mass-Duality>